Aayush Sood

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Education

University of Waterloo — Waterloo, ON
Honours Software Engineering, Co-op — *2026-2031*

Honours and Awards

- **Schulich Leader Scholarship (McGill)**: Winner of Canada's premier $100k STEM undergraduate scholarship (2026)
- **Scotiabank Software Engineering Scholarship (UWaterloo)**: $20k entrance scholarship for outstanding students (2026)
- **Wolf Scholar Finalist (UofT)**: National finalist for the Inaugural Wolf Scholarship (1 of 32 nationwide) (2026)
- **Lockheed Martin CYBERQUEST Competition**: Won 3rd Place in a team-based, global cybersecurity CTF challenge (2025)
- **Canadian Computing Competition Senior Division**: Certificate of Distinction (top 25%), 45/75 (2025)
- **Canadian Computing Competition Junior Division**: Certificate of Distinction and Honour Roll, ranked top 5%, 66/75 (2024)

Projects

IWKids - Emergency Room Triage App | *React Native, JavaScript, Expo, Git, OpenAI API* (GitHub)
- Built a mobile app that automates pediatric ER triage intake through real-time conversational support for patients and families, engineered with React Native and the OpenAI API
- Reduced simulated **triage data collection time by 40%**, saving triage nurses an estimated 1-2 hours per shift
- Delivered a working prototype in under 48 hours. Placed **2nd out of 24 teams** at the Atlantic Canada AI summit

CoinDeno - Crypto Tracker | *React.js, React Native, REST, Git, Firebase, CoinGecko APIs* (GitHub) (Link)
- Built and shipped a crypto tracker, letting users build custom portfolios, track live prices, and convert to local currency, growing an active user base to **20+ users** across the App Store, Google Play, and web
- Used Firebase to handle Google OAuth authentication and store user data (currency preferences, portfolio tickers), and integrated the CoinGecko API via REST for real-time pricing

SAP-1 - 6502 Microprocessor / 8-bit Project | *Python, Assembly* (GitHub: 6502) (GitHub: 8-bit)
- Built and assembled functional 8-bit and 6502 breadboard computers using 74LS-series discrete logic chips and SAP-1 architecture to execute hardware-level CPU operations
- Programmed low-level routines in Assembly to run working programs on the physical hardware, and debugged power, memory addressing, and signal issues directly on the breadboard

Market Maker | *Python* (GitHub)
- Designed a Python-based trading system simulation that dynamically adjusts buy/sell spreads and asset pricing based on real-time market volatility and automated inventory-level logic
- Compared PnL and inventory outcomes across 3 distinct trader profiles (Random, Momentum, and Informed) using NumPy and Matplotlib to visualize automated execution metrics

Experience

UWHPC (University of Waterloo High Performance Computing Team) - Core Developer — Oct 2026 – Present
- Joined as a core developer after completing the team's C++ performance onboarding challenge and a technical interview on the solution

University of Waterloo Engineering Society - Waterloo Engineering Competition Commissioner — Aug 2026 – Present
- Built and automated a sponsorship outreach pipeline using AI for company research and Google Apps Script to contact **940+ companies** for WEC Fall 2026, driving a **21x increase** in sponsorship discussions year-over-year

University of Waterloo Engineering Society - Career Fair Director — Jul 2026 – Present
- Led outreach to **45+ companies** spanning AI, quant finance, and engineering sectors to expand career fair employer coverage, securing registrations from regional target firms

Stanford University's Code in Place - Section Lead — May 2025, May 2026
- Selected to lead weekly Python instruction for two cohorts totaling **30+ global learners** as part of Stanford's Code in Place, a flagship global education initiative built on the CS106A (Intro to Programming) curriculum

Nova Scotia Secondary School Students' Association - IT Administrator — Apr 2025 – Jun 2026
- Maintained **100% ticket resolution** across M365, Entra, and Google Workspace for 100+ provincial users by authoring documentation and providing end-to-end admin support
- Architected a responsive redesign of www.nsssa.ca, modernizing the page architecture and asset loading to support registration and content delivery for hundreds of annual conference delegates

Skills

Languages: Python, C++, JavaScript, SQL
Frameworks: React, React Native, FastAPI, Node.js, HTML, CSS/SCSS, Tailwind, REST APIs
Tools: Firebase, Cloudflare, Netlify, Git, GitHub, Microsoft Azure, Linux
AI and Data Science: Claude Code, OpenAI API, scikit-learn, Pandas